CO2-AT1-Assignment1

Back

QUESTIONS

Q1
Smart Document Editor using String, StringBuilder and StringBuffer
10 marks

Q2
Hospital Token Management System using List Interface
10 marks

Q3
Airport Boarding Queue Management using Queue Interface
10 marks

Q5 **Smart Inventory Management using Map Interface** 10 marks

Smart Inventory Management using Map Interface

Store product information using an appropriate Map implementation.

Add new products.

Update product quantity.

Search a product using Product ID.

Calculate the total inventory value.

Display products category-wise

SIMATS Online Programming Lab

JAVA PROGRAMMING

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```
import java.util.Map;

public class Product {
    String name;
    String category;
```

5/5 answered

```
    double price;
    int quantity;

    public Product(String name, String categ
        this.name = name;
        this.category = category;
        this.price = price;
        this.quantity = quantity;
    }
    @Override
    public String toString() {
        return name + " | Category: " + cate
    }
    public static void main(String[] args) {
        Map<String, Product> inventory = new

        inventory.put("P101", new Product("L
        inventory.put("P102", new Product("S
        inventory.put("P103", new Product("D

        System.out.println("--- Initial Inve
        for (Map.Entry<String, Product> entr
            System.out.println("ID: " + entr
        }
        String updateId = "P101";
        if (inventory.containsKey(updateId))
            inventory.get(updateId).quantity
            System.out.println("\nUpdated qu
        }
        String searchId = "P102";
        System.out.println("\nSearching for
        System.out.println(inventory.getOrDe

        double totalValue = 0;
        for (Product p : inventory.values())
            totalValue += p.price * p.quanti
        }
        System.out.println("\nTotal Inventor

        System.out.println("\n--- Category-w
        for (Product p : inventory.values())
            if (p.category.equalsIgnoreCase(
                System.out.println(p);
            }
        }
```

```
        String removeId = "P103";
        inventory.remove(removeId);
        System.out.println("\nRemoved discon

        System.out.println("\n--- Final Avai
        for (Map.Entry<String, Product> entr
            System.out.println("ID: " + entr
        }
    }
}
```

Input

```
stdin input
```

Output

```
--- Final Available Inventory ---
ID: P102 -> Smartphone | Category:
Electronics | Price: $800.0 | Qty: 15
ID: P101 -> Laptop | Category: Electronics |
Price: $1200.0 | Qty: 12
```

Run Save